Unit 1, Lesson 1: Rewriting Numbers in Equivalent Forms

Benchmark

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

Learning Targets

- ☐ I can rewrite rational numbers in different but equivalent forms, including repeating decimals.
- ☐ I can explore patterns in fractions whose decimal equivalent is repeating.

1.1.1 Warm-Up: Day in the Life of a Mushroom Farmer

1. Alejandro, a mushroom-kit entrepreneur, says that when it comes to growing his business, “The most important thing is … adapting opportunities that come to you and still holding the same amount of passion.”

    a. What is something you are passionate about?

    b. How has a new interest or hobby surprised you?

1.1.2 Refresher: Rewriting Rational Numbers in Different Equivalent Forms

Rational numbers can be expressed as fractions, decimals, or percentages. Probabilities can be expressed using these forms as well.

1. Andrés and Jolie are discussing how to write $\frac{17}{20}$ as a decimal.

   Andrés says he can use long division to divide 17 by 20 to get the decimal.

   Jolie says she can write an equivalent fraction with a denominator of 100 by multiplying by $\frac{5}{5}$, then writing the number of hundredths as a decimal.

   a. Explain whether one, both, or neither of these strategies would work.

   b. Rewrite $\frac{17}{20}$ in its equivalent decimal form. Show your work.

2. Complete the statement.

   A percentage is a ratio expressed as a fraction with a denominator of _______.

3. There are many strategies that can be used to rewrite rational numbers in different forms. Some of them are included below.

   a. Work with your partner to match each phrase in the column on the left with a strategy in the column on the right. Draw arrows to make each match.

Rewriting Numbers in Different Forms	Strategy
To rewrite a fraction as a decimal, ...	... find an equivalent fraction with a denominator of 100.
To rewrite a decimal as a fraction, ...	... express it as a fraction out of 100 and simplify, if necessary.
To rewrite a fraction as a percentage, ...	... divide the numerator by the denominator.
To rewrite a percentage as a fraction, ...	... multiply or divide by 100.
To convert between decimals and percentages, ...	... use the place value of the digits to express the decimal as a fraction.

   b. Choose one phrase from the list on the left and complete it using a different strategy than what you matched it with above. Your strategy can be different from those listed above.

1.1.3 Guided Instruction: Repeating Decimals

A ***rational number*** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

1. Consider the fractions in the table shown.

   a. Rewrite each fraction in its equivalent decimal form. Then, indicate whether each decimal is a terminating or repeating decimal. If a decimal repeats digits in a pattern, use an ellipse (...) to indicate that the pattern continues.

Fraction Form	Decimal Form	Terminating or Repeating?
$\frac{1}{3}$		○ terminating ○ repeating
$\frac{3}{8}$		○ terminating ○ repeating
$\frac{13}{6}$		○ terminating ○ repeating
$\frac{38}{9}$		○ terminating ○ repeating
$\frac{17}{5}$		○ terminating ○ repeating

   b. Circle the fractions in the table that are greater than 1. Then, rewrite each as a mixed number.

   c. Based on your work in Part A, explain whether it is possible for a repeating decimal to ever terminate.

   d. Discuss with your partner any patterns you notice about the denominators of fractions whose decimal equivalent is repeating.

A number with a decimal that never ends but has a pattern that repeats infinitely is called a **repeating decimal.**

Repeating decimals are written with a bar over the digits that repeat. For example, the fraction $\frac{13}{6}$ is expressed as $2.1\overline{6}$ when written as a decimal because the 6 is the digit that repeats infinitely.

2. Isabel divided 85 by 99 to rewrite $\frac{85}{99}$ in decimal form. She found the resulting pattern $0.858585\ldots$ Write $\frac{85}{99}$ in its equivalent decimal form.

1.1.4 Exploration: Patterns in Repeating Decimals

1. Work with your partner to complete the following.

   a. Use a scientific calculator to find the decimal equivalent of each fraction.

Fraction	Decimal Equivalent
$\frac{1}{9}$	
$\frac{2}{9}$	
$\frac{3}{9}$	
$\frac{4}{9}$	
$\frac{5}{9}$	
$\frac{6}{9}$	

   b. Describe the relationship observed between fractions with a denominator of 9 and their decimal equivalents.

   c. Without performing any calculations, predict the decimal equivalent for each fraction in the table.

Fraction	Decimal Equivalent
$\frac{7}{9}$	
$3\frac{2}{9}$	
$17\frac{8}{9}$	

   d. Verify your predictions using your calculator.

2. Discuss with your partner how the values in the previous question could be used to determine the decimal equivalents for $\frac{1}{3}$ and $\frac{2}{3}$.

   a. Complete the equations.

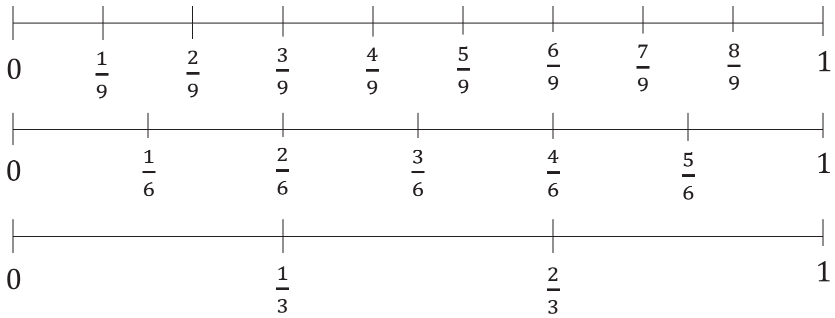
$$\frac{1}{3} = \frac{\boxed{}}{9}$$

$$\frac{2}{3} = \frac{\boxed{}}{9}$$

   b. Complete the table.

Fraction	Decimal Equivalent
$\frac{1}{3}$	
$\frac{2}{3}$	
$\frac{7}{3} = \boxed{}\frac{\boxed{}}{3}$	

3. Three number lines are shown.



a. Explain what the decimal equivalent of $\frac{2}{6}$ is using the number lines.

b. The decimal equivalent of $\frac{1}{6}$ is also a repeating decimal. Work with your partner to predict the decimal equivalent of $\frac{1}{6}$ based on the value of $\frac{2}{6}$.

c. Use a calculator to find the decimal equivalent of $\frac{1}{6}$. Write the value in decimal form.

d. Explain whether the decimal equivalent of $\frac{3}{6}$ is a repeating or terminating decimal.

1.1.5 Bring It Together: Patterns in Repeating Decimals

In the Exploration, you observed most fractions with denominators 3, 6, or 9 have a decimal equivalent that repeats.

1. Based on your observations, circle the fractions below that you predict also have repeating decimal equivalents.

$\frac{1}{11}$ $\frac{1}{12}$ $\frac{3}{12}$ $\frac{6}{15}$ $\frac{2}{15}$ $\frac{12}{99}$

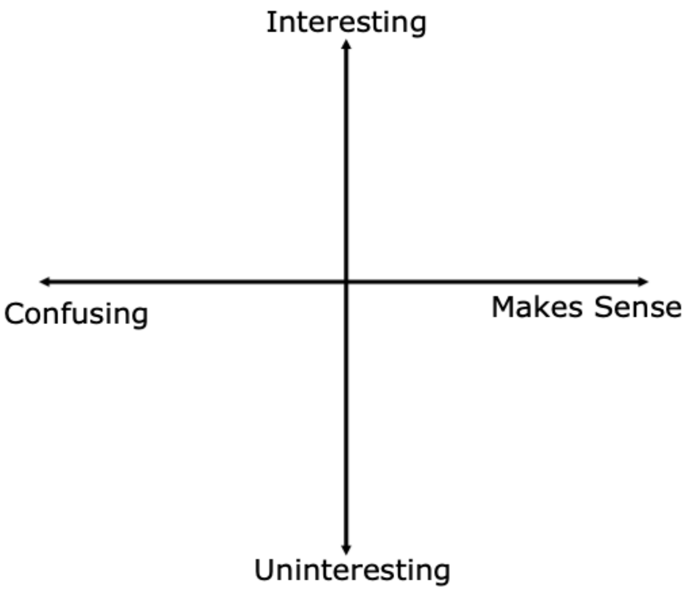
2. Use a scientific calculator to verify your predictions. Write the decimal equivalent for each fraction in the table.

Fraction	Decimal Equivalent
$\frac{1}{11}$	
$\frac{1}{12}$	
$\frac{3}{12}$	
$\frac{6}{15}$	
$\frac{2}{15}$	
$\frac{12}{99}$	

3. Make a conjecture about which fractions have decimal expansions that are repeating.

1.1.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you are on today's learning targets.



2. Write one question you still have about what you learned today.

Unit 1, Lesson 2: Converting Between Repeating Decimals and Fractions

Benchmark

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Learning Target

- [ ] I can rewrite rational numbers in different but equivalent forms, including repeating decimals.